

Bidirectional Self-Triggered Thyristor Crowbar

Bypass Protection of a Low-Impedance Element in the 3 kV / 3600 A Bench: Concept, Coordination and Functional Simscape Model

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Rev.	Date	Description
00	2026-09-12	First issue. Concept, coordination rules, worked example, Simscape component <code>crowbar_bidir</code> .

Abstract

An element that is structurally a short circuit (a few volts at the full 3600 A) sits in the current path of the 3 kV inductive test bench. If it opens, the loop inductance drives its terminal voltage up without limit. This note describes a bidirectional, fully passive thyristor crowbar placed in parallel with the element: two antiparallel thyristors, each fired by a floating break-over network, that short the element within a few microseconds in either polarity. The differences with respect to the unidirectional dump crowbar of the load are worked out: one thyristor is always forward biased in service, the voltage rise at opening is essentially unlimited, and the crowbar becomes a permanent short that must be cleared by the main switch. Coordination rules for the snubber capacitor, the MOV, the series reactor, the firing level and the thermal duty are given with a worked example, followed by the description of a functional Simscape component, `crowbar_bidir`, and a test procedure.

1 Purpose and scenario

The bench of Fig. 1 circulates $I_0 = 3600$ A in an RL load ($R = 0.8\ \Omega$, $L = 2$ mH) from a 3 kV source through a static switch S ; the load energy at opening is extracted by the dump crowbar described in `crowbar_thyristor_selftrigger`. A further element E is inserted in series in the loop. In service E is a low-impedance path: with $R_E \approx 1.4$ m Ω it drops about 5 V at full current. Should E open, whether by failure or by an unwanted command, the loop current has no path and the voltage across E rises at

$$\frac{dv_E}{dt} = \frac{I_0}{C_{par}}, \quad (1)$$

where C_{par} is whatever capacitance hangs on its terminals: with a few nanofarads this is hundreds of V/ns. The requirement is a device in parallel with E that turns into a short circuit as soon as v_E exceeds a threshold, in either polarity, without auxiliary supply, and stays shorted until the main switch clears the current.

Three features distinguish this crowbar from the dump crowbar of the load:

1. **Bias in service.** The dump thyristor is reverse biased at -3 kV all the time and sees forward voltage only when it must fire. Here $v_E \approx 5$ V, so one of the two antiparallel thyristors is *always* forward biased, at low voltage, and exposed to whatever transient E develops. Immunity to false firing is a design requirement, not a by-product.

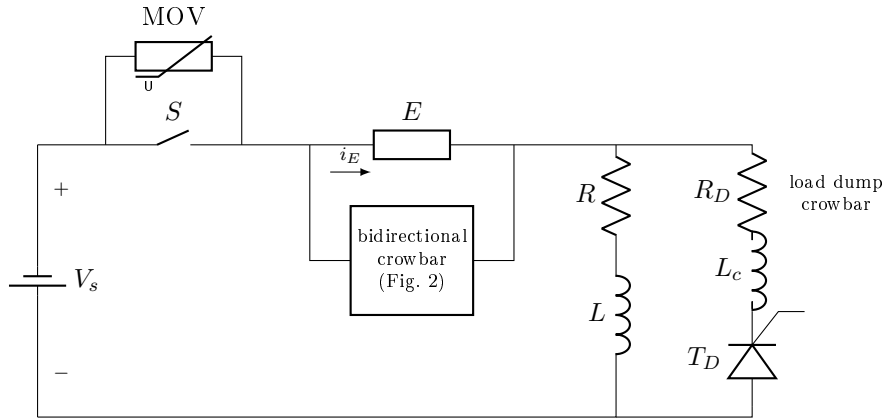


Figure 1: Bench with the low-impedance element E in the source branch and its bidirectional bypass crowbar. E may equally sit inside the load loop; the position changes the transient voltage seen in service (Section 3.3).

2. **Voltage rise.** At the dump crowbar the forward voltage is bounded by the MOV of the switch. Here nothing bounds v_E except what is deliberately placed across E : the time between the opening and the thyristor conduction (1–3 μ s) must be covered by a capacitor and a MOV.
3. **Duty after firing.** The dump crowbar carries a decaying current and turns off by itself. The bypass crowbar shorts E while the source keeps driving the loop: it carries I_0 continuously until S opens. It is a latch, and its firing must trip S .

2 Topology

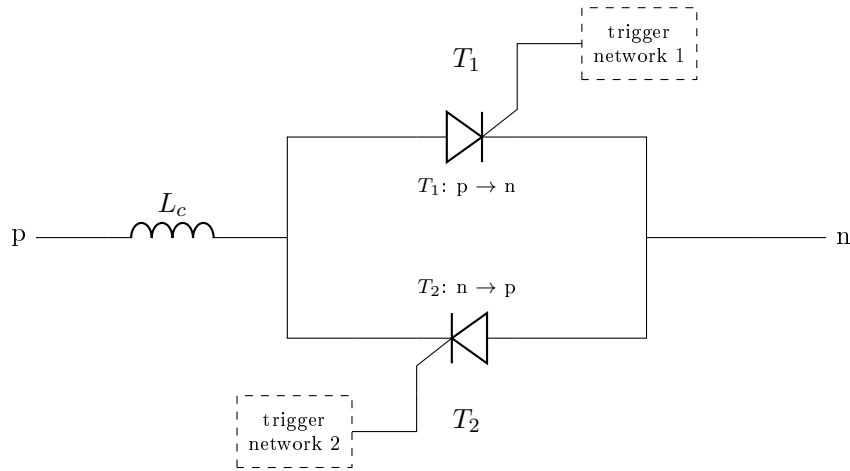


Figure 2: Bidirectional crowbar: two antiparallel thyristors with individual floating trigger networks and a small series reactor L_c that limits the di/dt of the current transfer. The RC snubber and the MOV are placed across the element (Fig. 1), not inside the crowbar branch. A bidirectional control thyristor (BCT, two thyristors in one press-pack) is an equivalent option.

For thyristor T_1 (conducting $p \rightarrow n$) the network is fed by $v = v_p - v_n$; for T_2 by $-v$. In service one of the two networks sees a forward voltage of a few volts, far below V_{arm} : D_1 and the arming TVS block, node N stays at cathode potential and the gate is quiet. The dynamic path C_g is retained only as an accelerator above V_{arm} ; the static BOD path is the primary trigger, because the quantity to detect is a voltage *level*, not a rate. The physics of both self-firing mechanisms

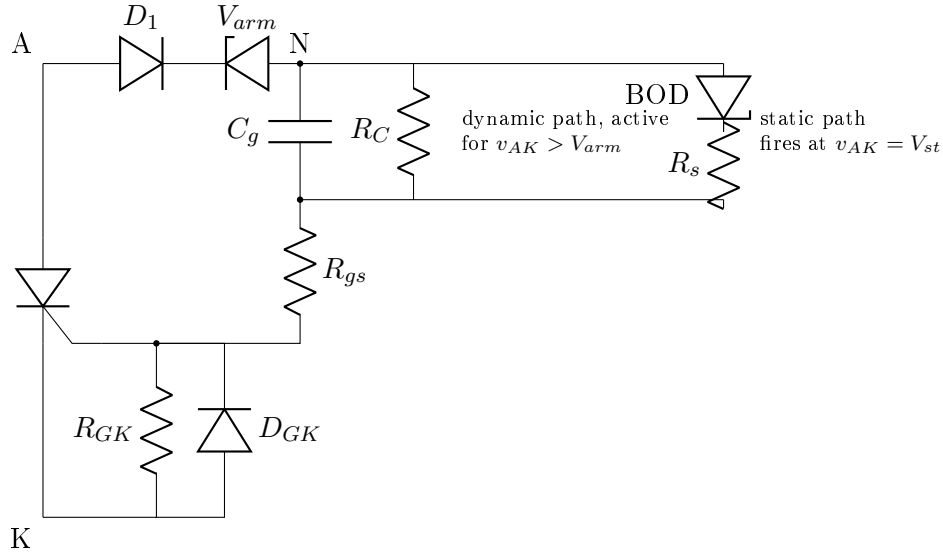


Figure 3: Trigger network of each thyristor, referred to its own cathode. Compared with the dump crowbar, an arming element of voltage V_{arm} (TVS or Zener stack) is added after D_1 : below V_{arm} no current reaches the gate, whatever the dv/dt . The static firing level is $V_{st} = V_{arm} + V_{BOD}$ (the BOD is drawn in its conduction direction).

and the reason for converting them into gate current are the same as for the dump crowbar and are not repeated here.

3 Sequence of events and coordination

3.1 The delay interval: snubber capacitor and MOV

Between the opening of E and the conduction of the thyristor, i.e. the turn-on delay t_{gd} (1–3 μ s for a large press-pack with hard gate drive) plus the time to reach V_{st} , the full I_0 must flow into something placed across E . Two elements share the task:

- An RC snubber C_{sn}, R_{sn} that turns (1) into a controlled ramp,

$$\frac{dv_E}{dt} \approx \frac{I_0}{C_{sn}}, \quad v_E(t) \approx R_{sn}I_0 + \frac{I_0}{C_{sn}}t. \quad (2)$$

C_{sn} is chosen so that the ramp stays below the critical dv/dt of the thyristor: the device must be turned on by its gate, not by displacement current, and the gate pulse is issued at V_{st} while the voltage continues to rise during t_{gd} . The resistive step $R_{sn}I_0$ appears instantly; if it exceeds V_{st} the crowbar is armed at $t = 0$, which is welcome, but $R_{sn}I_0$ also adds to the voltage reached at t_{gd} .

- A MOV in parallel, with knee above the voltage reached at t_{gd} in the nominal sequence and clamp level within the blocking capability of the thyristors. In the nominal sequence it carries little; it covers a longer delay, a missing gate pulse for a few tens of microseconds and the L_c overshoot. It cannot cover a thyristor that never fires: with the source connected, the energy at V_{MOV} grows as $V_{MOV}I_0t$ until S opens, tens of kilojoules in tens of milliseconds. Protection against a dead crowbar is the fast trip of S , Section 3.4.

3.2 Current transfer: the series reactor

Once the thyristor conducts, the current transfers from C_{sn} (and the MOV) into the thyristor branch. Without L_c the loop is the stray inductance of a few hundred nanohenry and the rate

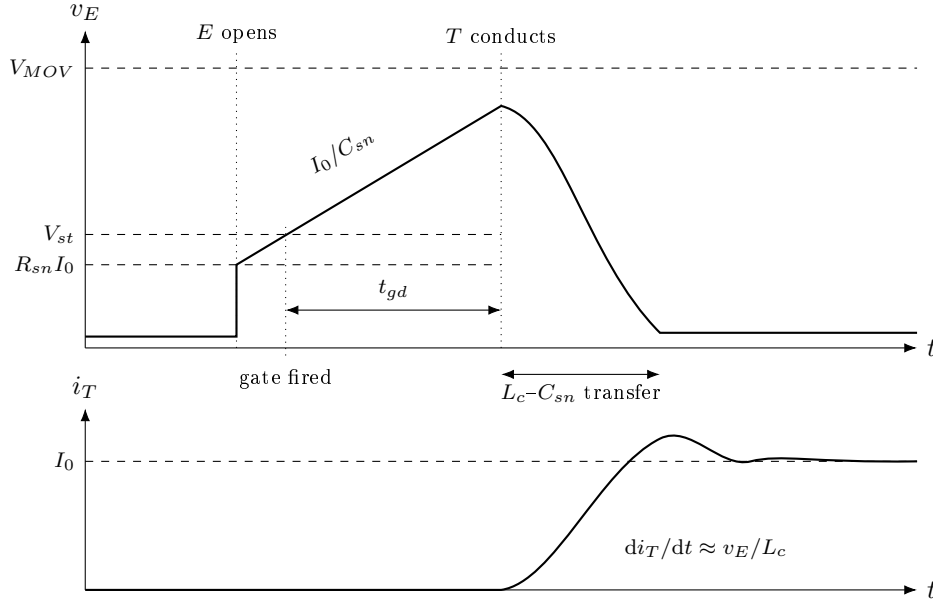


Figure 4: Qualitative sequence at the opening of E : resistive step of the snubber, linear rise at I_0/C_{sn} , gate fired at V_{st} , conduction after t_{gd} , transfer of the current into the thyristor through L_c with the L_c – C_{sn} overshoot. The MOV intervenes only if the thyristor is late.

of rise would be $v_E/L_\sigma \sim 1.4\text{ kV}/0.2\text{ }\mu\text{H} = 7\text{ kA}/\mu\text{s}$, far beyond any phase-control thyristor. A small reactor in the crowbar branch sets

$$\frac{di_T}{dt} \approx \frac{v_E(t_{gd})}{L_c}, \quad Z_0 = \sqrt{\frac{L_c}{C_{sn}}}, \quad \hat{i}_T \approx I_0 + \frac{v_E(t_{gd})}{Z_0}, \quad (3)$$

the last term being the L_c – C_{sn} exchange overshoot, lightly damped by R_{sn} ($\zeta = R_{sn}/2Z_0$). A saturable core is an option: it is a reactor during the transfer and disappears in continuous conduction. L_c costs nothing in service, since no current flows in the crowbar branch.

3.3 Static firing level versus normal transients

In service the forward-biased thyristor sees

$$v_E = R_E i_E + L_E \frac{di_E}{dt}, \quad (4)$$

and V_{arm} , hence V_{st} , must exceed the worst case of (4) with margin. The resistive term is a few volts. The inductive term depends on where E sits:

- *Inside the load loop* (in series with R , L): i_E is the load current, continuous through 2 mH; at the opening of S it changes at $I_0/\tau \approx 2\text{ A}/\mu\text{s}$ and the inductive term is negligible.
- *In the source branch* (in series with S , as drawn in Fig. 1): at the turn-off of S the current through E falls at the IGBT rate, kiloamperes per microsecond. With $L_E = 0.5\text{ }\mu\text{H}$ and $3\text{ kA}/\mu\text{s}$ the inductive term is 1.5 kV: a passive crowbar with V_{st} of a few hundred volts would fire at every opening of S . The options are a much higher V_{st} (with C_{sn} , MOV and thyristor class scaled accordingly), a very low-inductance E and connection, or giving up passive firing in favour of a controlled trigger through the external command input.

The same applies to any *intended* opening of E : a passive crowbar cannot be inhibited. If E is ever meant to open under current, this scheme does not apply.

3.4 Duty after firing and clearing

After firing, the crowbar is a short in series with the loop; the source keeps driving I_0 through it until S opens. Two consequences:

- The thyristor dissipates $P_T = (V_{T0} + r_T I_0) I_0$, about 7.8 kW at 3600 A for the reference device, continuously. With double-side cooling this is within its steady-state rating; for a bench that trips S within 100 ms the junction excursion is a few tens of kelvin and $\int i^2 dt \approx 1.3 \times 10^6 \text{ A}^2\text{s}$ is three orders of magnitude below the device limit. The reference device is oversized on voltage (it sees 5 V in service and 2 kV at most in the event) and adequate on current; a lower voltage class with the same wafer would do.
- The firing must trip S : from the model output M in simulation, and in hardware from a current transducer in the crowbar branch or from a voltage detector across E . Once S opens, the loop current goes to the dump crowbar of the load, the bypass current decays to zero and the thyristor turns off at I_H ; if E is in the source branch, no current flows through it at all after S opens.

3.5 Worked example

Reference values: $I_0 = 3600 \text{ A}$, $R_E \approx 1.4 \text{ m}\Omega$, thyristors 5STP 26N6500 ($V_{T0} = 1.12 \text{ V}$, $r_T = 0.29 \text{ m}\Omega$, critical dv/dt 1000 V/ μs , $I_{GT} \leq 0.4 \text{ A}$), $t_{gd} = 2 \mu\text{s}$.

Table 1: Worked example of the coordination.

Quantity	Value	Rule
V_{arm}	100 V	$\gg R_E I_0 + L_E di/dt$ in service (element in load loop)
V_{st}	400 V	$V_{arm} + V_{BOD}$; well below MOV knee
C_{sn}	8 μF	$I_0/C_{sn} = 450 \text{ V}/\mu\text{s} < \text{critical } dv/dt$
R_{sn}	0.15 Ω	step $R_{sn} I_0 = 540 \text{ V}$ (arms at $t = 0$); $\zeta = R_{sn}/2Z_0 \approx 0.1$
$v_E(t_{gd})$	1.44 kV	$R_{sn} I_0 + (I_0/C_{sn}) t_{gd}$
MOV	knee $\approx 1.6 \text{ kV}$, clamp $\approx 2 \text{ kV}$ at 3.6 kA	idle in the nominal sequence, back-up for a late gate
L_c	5 μH	$di_T/dt = 1.44 \text{ kV}/5 \mu\text{H} = 290 \text{ A}/\mu\text{s}$
$Z_0, \hat{i}_T$	0.79 Ω , $\approx 5.4 \text{ kA}$	overshoot $v_E/Z_0 \approx 1.8 \text{ kA}$ on top of I_0
EC_{sn}	8 J	$\frac{1}{2} C_{sn} v_E^2$, dissipated in R_{sn} and thyristor
P_T	7.8 kW	$(V_{T0} + r_T I_0) I_0$ until S opens
$\int i^2 dt$ (100 ms)	$1.3 \times 10^6 \text{ A}^2\text{s}$	vs. $\sim 2 \times 10^7 \text{ A}^2\text{s}$ of the device
Thyristor class	2.8–4.5 kV sufficient	$V_{RRM} \geq \text{MOV clamp with margin; } 6.5 \text{ kV reference oversized}$

With these values the voltage crosses V_{st} at $t = 0$ because of the resistive step, the thyristor conducts at 2 μs with 1.44 kV across it, and the current is in the thyristor 15 μs later. The MOV is a back-up. If C_{sn} is removed and only the MOV is kept, v_E reaches the clamp within nanoseconds at a rate that exceeds the critical dv/dt by orders of magnitude: the thyristor still turns on, but by displacement current over a random area, and the model flags it with $|M| = 4$.

4 Functional Simscape component crowbar_bidir

4.1 Interface

Two electrical conserving ports **p** (left) and **n** (right); T_1 conducts $p \rightarrow n$, T_2 conducts $n \rightarrow p$. Physical-signal input **G** (bottom): external fire command, dimensionless, fires the forward-biased

thyristor when above 0.5; connect a PS Constant 0 if unused. Physical-signal output **M** (top): signed state code, zero while blocking, sign equal to the conduction direction (+1 for T_1), magnitude equal to the trigger source of Table 2.

Table 2: Trigger source codes reported on $|M|$ (priority order).

$ M $	Source	Meaning
1	static (BOD)	gate current reached I_{GT} with the BOD conducting: nominal firing at V_{st}
2	dynamic (C_g)	gate current reached I_{GT} through C_g alone, above V_{arm}
3	external	command $G > 0.5$
4	intrinsic dv/dt	dv/dt exceeded the critical value with $v > V_{min}$ before any gate path: uncontrolled turn-on, increase C_{sn} or lower V_{st}
5	breakover	v exceeded V_{BO} before any gate path: uncontrolled turn-on

4.2 Equations

Main branch, with i positive p→n and $d \in \{+1, -1\}$ the conduction direction:

$$\text{blocking: } i = G_{off} v, \quad (5)$$

$$\text{conducting: } v = d V_{T0} + r_T i. \quad (6)$$

Trigger networks ($k = 1$ for T_1 with $v_1 = v$, $k = 2$ for T_2 with $v_2 = -v$):

$$i_{d,k} = \begin{cases} (v_k - V_{arm} - v_{C,k})/R_{ser} & v_k - V_{arm} - v_{C,k} > 0 \\ (v_k - V_{arm} - v_{C,k}) G_{D1,off} & \text{otherwise} \end{cases} \quad (7)$$

$$C_g \frac{dv_{C,k}}{dt} = i_{d,k} - \frac{v_{C,k}}{R_C} - i_{st,k}, \quad i_{st,k} = \frac{\max(v_{C,k} - V_{BOD}, 0)}{R_s}, \quad V_{BOD} = V_{st} - V_{arm}, \quad (8)$$

$$v_{gk,k} = V_{GT} \tanh\left(\frac{i_{d,k} R_{GK}}{V_{GT}}\right), \quad i_{j,k} = i_{d,k} - \frac{v_{gk,k}}{R_{GK}}. \quad (9)$$

The junction current $i_{j,k}$ reaches I_{GT} when $i_{d,k} = I_{GT} + V_{GT}/R_{GK}$. The terminal current is $i_t = i + i_{d,1} - i_{d,2}$. The rate of change of v is estimated with a first-order filter of time constant τ_f , so that no derivative appears in event conditions. Two integrators, $\int i^2 dt$ and $\int v i dt$, are exposed for the thermal check of Section 3.4.

4.3 State machine

Five event variables: **on**, **dir** (d), **armed**, **latched**, **mode**, plus the trigger time **t_trig**.

1. **Arming.** While blocking and not armed, the first of the conditions of Table 2 that becomes true for T_1 ($v > 0$) or T_2 ($v < 0$) sets **armed** = 1, **dir**, **mode** and records the time.
2. **Turn-on.** At **t_trig** + t_{gd} : if $dv > 0$ still holds, **on** = 1; otherwise the arming is cancelled (polarity reversed during the delay).
3. **Latching.** **latched** = 1 when $di > I_L$.
4. **Turn-off.** **on** = 0 when $di < I_H$ after latching. The latch flag prevents a spurious turn-off at the switching instant, when the current starts from zero.

All trigger sources, including the intrinsic ones, go through the same delay: the code on M records which condition came first, which is the design information needed.

4.4 Parameters

Table 3: Parameters and defaults of `crowbar_bidir`.

Group	Name	Default	Meaning
On-state	Vt0, rT	1.12 V, 0.29 m Ω	PWL on-state
	Goff	1×10^{-6} S	off-state conductance, both polarities
	I1, Ih	2 A, 1 A	latching, holding current
	tgd	2 μ s	turn-on delay (> 0)
Gate	Igt, Vgt, Rgk	0.4 A, 3 V, 10 Ω	trigger current/voltage, R_{GK}
Network	Varm, Vst	100 V, 400 V	arming and static firing levels
	Cg, Rc	1 nF, 22 k Ω	dynamic path, reset
	Rs, Rser	100 Ω , 11 Ω	static path resistor, loop series resistance
	Gdloff	1×10^{-9} S	D_1 off-state conductance
Intrinsic	dvd_t_crit, Vmin	1×10^9 V/s, 50 V	intrinsic dv/dt firing
	Vbo, tau_f	7800 V, 50 ns	breakover, dv/dt sensing

Not modelled: plasma spreading and di/dt limits, reverse recovery, t_q , thermal feedback on V_{T0} and r_T . The reactor, the snubber and the MOV are external blocks, so that their values can be swept.

5 Simulation test procedure

Model E as a resistor R_E in series with an ideal switch that opens at t_0 with the full I_0 established, a stray capacitance C_{par} of a few nanofarad across it, the RC snubber, the MOV (the existing varistor or `tvsc_lamp` model), and the crowbar branch L_c + `crowbar_bidir`. Fixed step 10 ns is adequate ($C_g R_{ser} \approx 10$ ns, $\tau_f = 50$ ns).

1. **Nominal event.** Record v_E , i_T , M , the MOV current and $i_{j,1}$. Expected: arming at t_0 (resistive step), $M = +1$ at $t_0 + t_{gd}$, peak v_E per Table 1, di_T/dt and overshoot per (3), MOV current negligible.
2. **Snubber removed.** Keep the MOV only: M should read +4, the sign that the design relies on displacement-current turn-on. Reinstate C_{sn} and find the minimum value that restores $M = +1$; compare with $I_0/(dv/dt)_{crit}$.
3. **Delay sweep.** t_{gd} from 1–5 μ s: $v_E(t_{gd})$ must stay below the MOV knee; note where the MOV starts to share current.
4. **Reactor sweep.** L_c from 1–10 μ H: read di_T/dt in the first microseconds and the overshoot against the device rating.
5. **Reverse polarity.** Repeat with the current reversed (or E in the other orientation): $M = -1$, symmetric waveforms.
6. **False-firing margin.** With E closed, apply the normal bench sequence (closing and opening of S , firing of the dump crowbar) and check that $\max|v_E| < V_{arm}$ with margin and that M stays 0; if E is in the source branch, include the element inductance L_E explicitly.

7. **Clearing.** After firing, open S per the bench logic and verify that i_T decays to zero, on drops at I_H and i_{2t} , E_d match the thermal estimate.

6 Conclusions

A bidirectional, passively fired thyristor crowbar is a sound way to protect a structurally short element against an unwanted opening at 3600 A: it needs no auxiliary power, fires within microseconds in either polarity and carries the full current until the main switch clears. Its design differs from the dump crowbar in three points that the coordination of Section 3 addresses: an arming threshold well above the service transients of the element (whose position in the loop decides how large they are), a snubber capacitor and MOV that hold the voltage during the turn-on delay while keeping the dv/dt below the critical value, and a small series reactor that keeps the current transfer within the di/dt capability of the device. The crowbar is a latch: its firing must trip the main switch, and it cannot be inhibited, so it is only applicable to an element that is never meant to open under current. The functional component `crowbar_bidir` reproduces the trigger paths, the delay and the latch, and reports on M which mechanism fired first, which is the quantity to watch when sizing C_{sn} and V_{st} .

References

- [1] D. Bagnara, “Crowbar thyristor for the 3 kV / 3600 A static-switch test bench: device selection and passive self-triggering gate circuit,” technical note `crowbar_thyristor_selftrigger`, Rev. 00, 2026.
- [2] Hitachi Energy (formerly ABB Semiconductors), “Phase Control Thyristor 5STP 26N6500,” data sheet.
- [3] B. J. Baliga, *Fundamentals of Power Semiconductor Devices*. New York: Springer, 2008, ch. 8 (Thyristors).
- [4] “Protective circuitry for a thyristor,” U.S. Patent 4 651 251 (break-over diode protective firing).
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